Workshop #3

Regression & Prediction

Appalachian A. I. Corps @ UTK

Notes

Lesson Objective

In this lesson, you will learn how to use scatterplots and regression lines in Python to predict nitrate levels from RGB color values on test strips.

Materials Needed:

A computer with a webcam
A web browser (Chrome, Firefox, or Safari)

Notes

Workshop Structure

Navigate to:

https://tinyurl.com/aaic-wq-3

Notes

Review

Last workshop we learned:

- AI: Classification
  - Model Accuracy
- Agricultural Uses of Classification
- Data Ownership
- Computers & Color

Notes

Section 1

Quick Catch Up: Computers & Color Activity

Notes

RGB Values –> Color

Checkpoint 7. a: Using RGB Values to Create Color

Materials Needed:

- Slider Tool on Pg. 0
- Handout from last workshop

Notes

Using RGB Values to Create Color

Use the slider tool below to convert the RGB values (from last workshop) to colors
Record your answers on the handout
We'll do one together!

Notes

Using RGB Values to Create Color

Checkpoint 7. b: Use the slider tool!
What color does this RGB value represent? (110, 164, 212)
Just like the computer last workshop, try to classify/make a prediction about what fruit this could be!

Notes

Section 2

Workshop #3

Notes

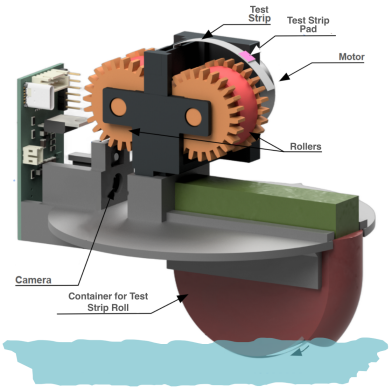
Remember Me, Smokey Buoy!?

We met Smokey Buoy last workshop and learned a bit about how it works.

Notes

Smokey Buoy: Components

Let's recall Smokey's components and how they work together



Notes

Smokey Buoy: Camera

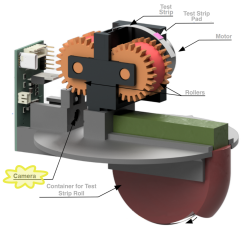
Smokey Buoy's camera has two jobs:

Watch for a test pad

- We talked about this last workshop!
- *Recall:* What is the process called?

When pad is present, take a picture

- We'll discuss what happens with the picture today!
- *A new process!*



Notes

Section 3

Introduction to Regression

Notes

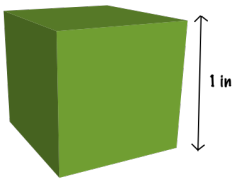
Introduction to Regression

Last workshop, we learned about **classification**.
In this workshop, we will learn about **regression**.
Regression is all about **predicting** something (a number!) from something else.

Notes

Scenario 1: Cube Towers

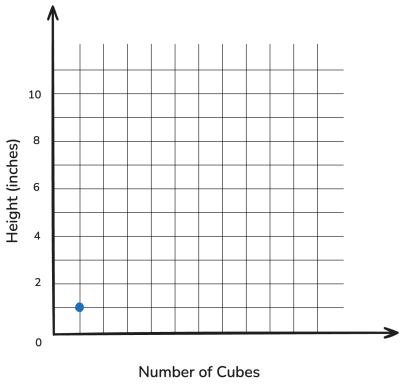
We want to build a tower
We have 1" inch cubes to use



Notes

Let's Predict! — 1 Cube

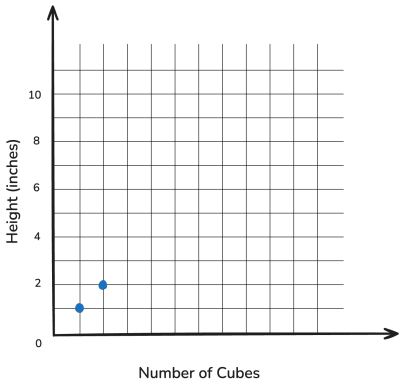
# of Cubes	Height (inches)
1	1



Notes

Let's Predict! — 2 Cubes

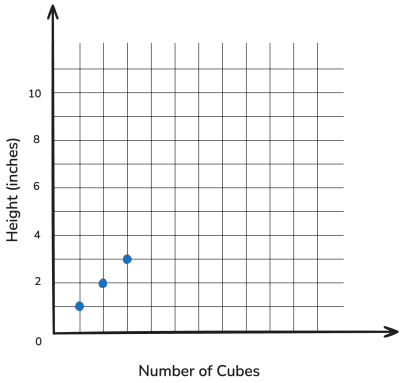
# of Cubes	Height (inches)
1	1
2	2



Notes

Let's Predict! — 3 Cubes

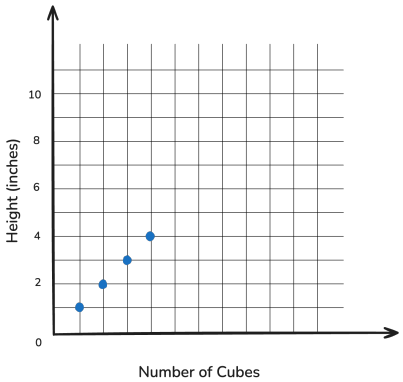
# of Cubes	Height (inches)
1	1
2	2
3	3



Notes

Let's Predict! — 4 Cubes

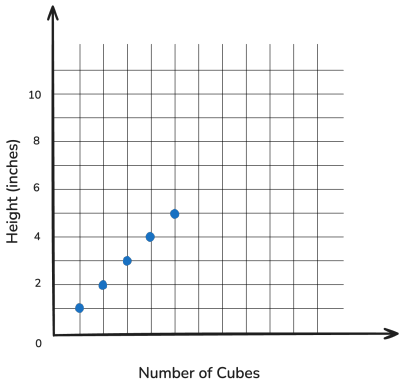
# of Cubes	Height (inches)
1	1
2	2
3	3
4	4



Notes

Let's Predict! — 5 Cubes

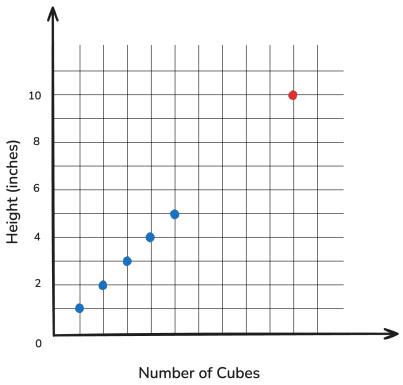
# of Cubes	Height (inches)
1	1
2	2
3	3
4	4
5	5



Notes

Let's Predict! — 10 Cubes

# of Cubes	Height (inches)
1	1
2	2
3	3
4	4
5	5
10	10



Notes

Discuss: In Your Group

Checkpoint 2. a: How tall will our tower be if we use **100 cubes**? How do you know?

Notes

Scenario 2: Cheap Uncle Bob

We want to build a much taller tower, but we ran out of 1" cubes. Uncle Bob volunteers to make us more cubes — cheaper!

Notes

Scenario 2: Cheap Uncle Bob

Oh no! Uncle Bob's cubes are all slightly different heights!

Notes

Discuss: In Your Group

Checkpoint 2. b: How will Uncle Bob's new cubes affect our tower height predictions?

Notes

Section 4

The Case for Regression

Notes

The Case for Regression

Bob's cubes saved us money, but now tower heights are all over the place. We need a better way to predict.

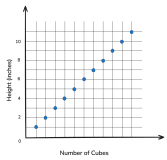


Figure 1: True with perfectly 1" cubes

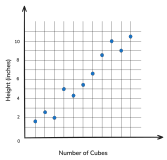


Figure 2: Reality with Uncle Bob's cubes

Notes

How Do We Fit Our Line?

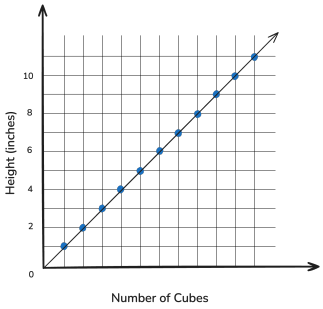


Figure 3: Very clear line with perfect cubes

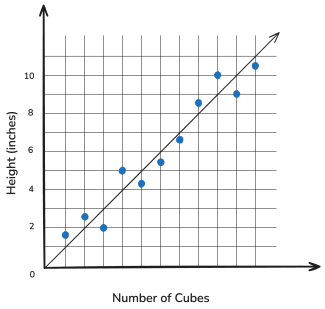


Figure 4: Need **regression** — the mathematical best-fit line

Notes

Smokey Buoy: Regression

The buoy also uses **regression** to predict nitrate levels.
Let's learn a bit more about this!

Notes

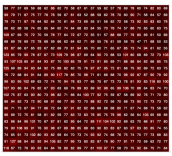
Section 5

Regression Activity — Pt. I:RGB and Scatterplots

Notes

Meaning RGB

In Workshop #2, we learned how computers "see" color through RGB values. The buoy takes **all the pixels** for each test pad and finds their **mean**.



88	89	92	92	93
87	84	87	90	93
89	81	84	85	88
77	80	81	90	85
71	77	77	81	88

Notes

RGB Activity (Part 1)

Materials Needed:

RGB Applet V2
Deck of Purple "Nitrate Test Pad" Cards
Pencil / Highlighter (optional)
Handout / Dot Stickers
Large Graph Paper / Poster Markers

Notes

RGB Activity (Part 1) — Instructions

Open the RGB Applet V2 in a new tab.
Find card #1 in your deck and hold it in front of your webcam.
Using the middle square in the frame, record the RGB values on your handout.
Repeat for all other numbered cards in your deck.

Notes

Scatterplots Activity (Part 2)

Notes

Scatterplots Activity (Part 2) — Instructions

This portion will be completed with your group. Each group is assigned either **red**, **green**, or **blue**.

Notes

Scatterplots Activity (Part 2) — Instructions

Using the round stickies and graph paper, create a scatterplot.
Place color values (R, G, or B — per your group) on the **x-axis**.
Place nitrate concentrations on the **y-axis**.

Notes

Section 6

Regression Activity — Pt. 2:Regression Modeling

Notes

Regression Modeling

Notes

Storing Your Data

Replace the ?? placeholders with your color values and nitrate concentrations.

IMPORTANT: Values must be in the **exact same order**.

Click **Run Code** to store your data.

```
color_data = [188, 179, 176, 176, 172, 167, 166, 162]
nitrate_ppm = [0, 1, 2, 3, 4, 5, 6, 7]

print(color_data)
print(nitrate_ppm)

color_data, nitrate_ppm = process_data(color_data, nitrate_ppm)
df = process_df(color_data, nitrate_ppm)
```

```
[188, 179, 176, 176, 172, 167, 166, 162]
[0, 1, 2, 3, 4, 5, 6, 7]
```

Notes

Recreating Your Team's Scatterplot

Replace the ?? placeholders:

First blank → x variable: `color_data`
Second blank → y variable: `nitrate_ppm`

Click **Run Code** to create a scatterplot matching your group's paper plot.

```
plt.figure(figsize=(6, 5))
plt.scatter(df["color_data"], df["nitrate_ppm"], s=60, color="black")
plt.title("Nitrate vs. Color")
plt.xlabel("Color")
plt.ylabel("Nitrate (ppm)")
plt.tight_layout()
plt.show()
```

Notes

Running the Linear Regression

Replace the ?? placeholders:

First blank → `color_data`
Second blank → `nitrate_ppm`

Click **Run Code** to generate the line of best fit equation.

```
slope, intercept = linear_regression(color_data, nitrate_ppm)

print("slope (m):", slope)
print("intercept (b):", intercept)
```

```
slope (m): -0.28630278063851533
intercept (b): 53.10195674562279
```

Notes

From Regression to Equation

The linear regression produces the equation for the **line of best fit**.

Remember $y = mx + b$? We can just substitute!

Click **Run Code** to see the slope-intercept equation.

```
print(f"y = {slope:.2f}x + {intercept:.2f}")
```

```
y = -0.29x + 53.10
```

Notes

Plotting the Regression Line

Click **Run Code** to plot the scatterplot with the regression line.

```
nitrate_pred = slope * color_data + intercept

plt.figure(figsize=(6, 5))
plt.scatter(color_data, nitrate_ppm, s=60, color="black")
plt.plot(color_data, nitrate_pred, color="black", linewidth=1)
plt.title("Nitrate Concentration vs. Color")
plt.xlabel("Color")
plt.ylabel("Nitrate Concentration (ppm)")
plt.text(
    min(color_data), max(nitrate_ppm),
    f"y = {slope:.2f}x + {intercept:.2f}\n",
    ha="left", fontsize=10, color="gray"
)
plt.tight_layout()
plt.show()
```

Notes

Activity — Draw Your Line of Best Fit

Materials Needed:

Poster Markers
Scatterplot
Yardstick

Notes

Activity — Draw Your Line of Best Fit

Using the exact positioning Python generated, **copy the regression line onto your group's scatterplot**. When finished, bring your group's plot to the front of the class.

Notes

Section 7

Prediction & R^2

Notes

Making Predictions — Reload Your Data

Re-enter your color values and nitrate concentrations.

NOTE: You can copy/paste the two lists from the previous activity.

Click **Run Code** to re-load your data.

```
color_data = [188, 179, 176, 176, 172, 167, 166, 162]
nitrate_ppm = [0, 1, 2, 3, 4, 5, 6, 7]

# Do not change the code below -----------------------------------
color_data, nitrate_ppm = process_data(color_data, nitrate_ppm)
slope, intercept = linear_regression(color_data, nitrate_ppm)
print(f"Slope: {slope:.2f}, Intercept: {intercept:.2f}")
```

```
Slope: -0.29, Intercept: 53.10
```

Notes

Using the Model to Predict

Now that we have our regression model, we can **predict** nitrate concentrations for new color values — even ones we didn't measure!

Click **Run Code**, then type any color value into the box to see the predicted nitrate concentration.

```
show_prediction_plot(color_data, nitrate_ppm, slope, intercept)
```

Notes

Predict Nitrate Concentrations for New Test Pads

Materials Needed:

RGB Applet V2
Three Additional "Nitrate Test Pad" Cards
Handout / Pencil

Notes

Predict Nitrate Concentrations for New Test Pads

Use the applet to find the R, G, or B value for the new cards (per your group).
Use the graphing tool to **predict the nitrate concentration (ppm)**.
Record these predictions on your handout.

Notes

How Good Is Our Model? R^2

R^2 (R-squared) tells us how well our regression line fits the data — like a report card:

R^2 value	What it means
Close to **100%**	Line fits very well — predictions will be accurate
Around **70–90%**	A reasonably strong fit
Close to **0%**	Line barely fits — predictions won't be reliable

Notes

Calculate Your R^2 Value

Click **Run Code** to calculate your model's R^2 value.

```
r_squared = calculate_rsquared(color_data, nitrate_ppm, slope, intercept)

print(f"R² = {r_squared * 100:.1f}%")
```

```
R² = 94.8%
```

Notes

Discuss: In Your Group

Checkpoint 6. a: Write your group's R^2 value on the poster at the front of the room.

Compare with the other groups — what do you notice? **How might we improve our model and, therefore, our predictions?**

Notes

Section 8

Exit Ticket

Notes

Exit Ticket

Great job! You've learned so much!

Share what you've learned on the **Exit Ticket**.

Notes

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